181. Steel becomes soft and pliable by

(a) Annealing

(b) Nitriding

(c) Tempering

(d) Case hardening

Answer: A — (a) Annealing

Chemical reason: Annealing makes steel softer and more pliable by heating it and allowing slow cooling, which reduces hardness and internal stresses.

182. Most stable oxidation state of iron is

(a) +2

(b) +3

(c) -2

(d) -3

Answer: B — (b) +3

Chemical reason: The result is obtained by assigning oxidation numbers before and after the reaction and balancing electron transfer. The species in +3 has the oxidation-state change required by the stated redox conditions, so that choice is correct.

183. Nickel steel contain % of Ni

(a) 1-5%

(b) 3-5%

(c) 6-5%

(d) 8-5%

Answer: B — (b) 3-5%

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in 3-5% matches the relevant metallurgical process or alloy property.

184. Permanent magnet is made from

(a) Cast iron

(b) Steel

(c) Wrought Iron

(d) All of these

Answer: B — (b) Steel

Chemical reason: Permanent magnets require a material that can retain magnetization. Hardened steel has high coercivity and remanence, so it is used for permanent magnets.

185. In nitriding process of steel

(a) Steel is heated in an atmosphere of ammonia

(b) Steel is made red hot and then cooled

(c) Steel is made red hot and then plunged into oil for cooling

(d) None of these

Answer: A — (a) Steel is heated in an atmosphere of ammonia

Chemical reason: In nitriding, steel is heated in an atmosphere of ammonia. Nitrogen diffuses into the surface and forms hard nitrides, producing a wear-resistant case.

186. Iron on reacting with carbon give

(a) FeC
(b) Fe_2C
(c) Fe_3C
(d) FeC_2

Answer: C — (c) Fe_3C

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Fe_3C matches the relevant metallurgical process or alloy property.

187. Iron loses magnetic property at

(a) Melting point
(b) 1000K
(c) Curie point
(d) Boiling point

Answer: C — (c) Curie point

Chemical reason: Iron loses ferromagnetism above its Curie temperature because thermal agitation destroys long-range alignment of magnetic domains.

188. Heat treatment alters the properties of steel due to

(a) Chemical reaction on heating
(b) Partial rusting
(c) Change in the residual energy
(d) Change in the lattice structure due to differential rate of cooling

Answer: D — (d) Change in the lattice structure due to differential rate of cooling

Chemical reason: Heat treatment changes steel because different cooling rates produce different crystal/microstructural forms (for example pearlite or martensite), which strongly alter hardness and toughness.

189. Pure conc. HNO_3makes iron passive as the surface is covered with protective layer of

(a) Fe_2O_3
(b) FeO
(c) Fe_3O_4
(d) $Fe(NO_3)_3$

Answer: C — (c) Fe_3O_4

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Fe_3O_4 matches the relevant metallurgical process or alloy property.

190. Red hot iron absorbs SO_2 giving the product

(a) $FeS+O_2$
(b) Fe_2O_3+FeS
(c) FeO+FeS
(d) FeO+S

Answer: C — (c) FeO+FeS

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in FeO+FeS matches the relevant metallurgical process or alloy property.

191. If steel is heated to a temperature well below red heat and is then cooled slowly, the process is called

(a) Tempering

(b) Hardening

(c) Softening

(d) Annealing

Answer: A — (a) Tempering

Chemical reason: Heating hardened steel to a temperature below red heat and then cooling it is tempering. Tempering reduces brittleness while retaining useful hardness.

192. In smelting of iron, which of the following reactions takes place in Blast furnace at 400oC-600oC

(a) $CaO+SiO_2 \rightarrow CaSiO_3$

(b) $2FeS+_3O_2 \rightarrow {}_2Fe+SO_2$

(c) $FeO+SiO_2 \rightarrow FeSiO_3$

(d) $Fe_2O_3+_3CO \rightarrow {}_2Fe+_3CO_2$

Answer: D — (d) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ matches the relevant metallurgical process or alloy property.

193. Soil containing both Al and Fe is called

(a) Laterite

(b) Bauxite

(c) Pedalfers

(d) Clay

Answer: C — (c) Pedalfers

Chemical reason: Lateritic/pedalferric soils are rich in iron and aluminium oxides after intense leaching. The option given identifies soil containing both Al and Fe.

194. German silver is an alloy of

(a) Copper, zinc and nickel

(b) Copper and silver

(c) Copper, zinc and tin

(d) Copper, zinc and silver

Answer: A — (a) Copper, zinc and nickel

Chemical reason: German silver is a Cu–Zn–Ni alloy. Despite the name, it contains no silver.

195. Iron is rendered passive by the action of

(a) Conc. H_2SO_4

(b) Conc. H_3PO_4

(c) Conc. HCl

(d) Conc. HNO_3

Answer: D — (d) Conc. HNO_3

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Conc. HNO_3 matches the relevant metallurgical process or alloy property.

196. Iron sheets are galvanized by depositing a coating of or In galvanisation, iron surface is coated with

(a) Zinc
(b) Tin
(c) Chromium
(d) Nickel

Answer: A — (a) Zinc

Chemical reason: Galvanisation protects iron by coating it with zinc. Zinc provides both a physical barrier and sacrificial protection because Zn oxidizes more readily than Fe.

197. Chemical formula of rust is

(a) FeO
(b) Fe_3O_4
(c) $Fe_2O_3.xH_2O$
(d) $FeO.xH_2O$

Answer: C — (c) $Fe_2O_3.xH_2O$

Chemical reason: This is a standard nomenclature/composition identification: the name or formula given in the question corresponds to $Fe_2O_3.xH_2O$. The choice is fixed by the known composition of that compound/mineral.

198. Heating steel to bright redness and then cooling suddenly by plunging it into oil or water, makes it

(a) Hard and pliable
(b) Soft and pliable
(c) Soft and brittle
(d) Hard and brittle

Answer: D — (d) Hard and brittle

Chemical reason: Quenching red-hot steel rapidly in water or oil forms hard martensitic structures. The steel becomes very hard but also brittle.

199. Which of the following is found in body

(a) Pb
(b) Fe
(c) Cd
(d) Al

Answer: B — (b) Fe

Chemical reason: The correct choice is Fe. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

200. Which of the following pairs of elements might form an alloy

(a) Zinc and lead

(b) Iron and mercury

(c) Iron and carbon

(d) Mercury and platinum

Answer: C — (c) Iron and carbon

Chemical reason: Iron and carbon form a wide range of Fe–C alloys, including steel and cast iron. The other listed pairs do not form the standard structural alloy intended by the question.